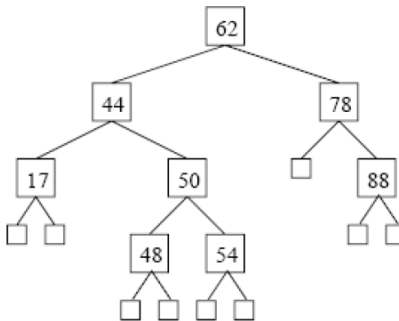


INT202 Complexity of Algorithms

Tutorial 3

Q1. Show that array [20 15 18 7 9 5 12 3 6 2] forms a max-heap.

Q2. Draw the AVL tree resulting from the insertion of an item with a key 52 into the given binary tree. (Only the final tree is needed.)



Q3. Draw the binary min-heap that results from inserting: 77, 22, 9, 68, 16, 34, 13, 8 in that order into an initially empty binary min heap. You are required to draw the intermediate heaps after each insertion and the final heap.

Q4. Insert the following sequence of elements into an AVL tree, starting with an empty tree: 10, 20, 15, 25, 30, 16, 18, 19. Draw the AVL tree after each insertion.